THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 4: **Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Using $a^2 - b^2$, the value of 47×53 is: 1

(a) 2492 **(b)** 2490 **(c)** 2493 **(d)** 2491

2. $x^2 - 81$ factorises as: 1

(a) $(x - 9)^2$ **(b)** $(x + 9)^2$ **(c)** $(x + 81)(x - 1)$ **(d)** $(x + 9)(x - 9)$

3. If $a = 4$ and $b = 9$, then $(a + b)^2 =$ 1

(a) 171 **(b)** 169 **(c)** 170 **(d)** 168

4. $(x + 3)(x + 7) =$ 1

(a) $x^2 + 10x - 21$ **(b)** $x^2 + 10x + 21$ **(c)** $x^2 + 4x + 21$ **(d)** $x^2 + 11x + 21$

5. $(x + 6)^2 =$ 1

(a) $x^2 + 36$ **(b)** $x^2 + 6x + 36$ **(c)** $x^2 + 12x + 36$ **(d)** $x^2 - 12x + 36$

6. $(x - 9)^2 =$ 1

(a) $x^2 - 18x + 81$ **(b)** $x^2 + 81$ **(c)** $x^2 + 18x + 81$ **(d)** $x^2 - 9x + 81$

7. $(x + 10)(x - 10) =$ 1

(a) $x^2 + 20x - 100$ **(b)** $x^2 - 100$ **(c)** $x^2 + 100$ **(d)** $x^2 - 20x + 100$

8. Using $(a + b)^2$, the value of 61^2 is: 1

(a) 3720 **(b)** 3722 **(c)** 3723 **(d)** 3721

9. Using $a^2 - b^2$, the value of 46×54 is: 1

(a) 2484 **(b)** 2486 **(c)** 2483 **(d)** 2485

10. $x^2 - 121$ factorises as: 1

(a) $(x + 121)(x - 1)$ **(b)** $(x + 11)^2$ **(c)** $(x - 11)^2$ **(d)** $(x + 11)(x - 11)$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find 64^2 using $(a + b)^2$. 2

12. Expand $(x + 4)^2$. 2

13. Expand $(x + 2)(x - 2)$. 2

14. Find 82^2 using $(a + b)^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Expand $(2x + 7)^2$. 3

16. Factorise $x^2 + 14x + 49$. 3

17. Evaluate 59×41 using an identity. 3

OR

Expand $(3x + 4)^2$. 3

18. Factorise $x^2 + 18x + 81$. 3

OR

Evaluate 60×40 using an identity. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Expand $(4x + 3y)^2$ and find its value when $x = 1$ and $y = 1$. 5

OR

Factorise $x^2 + 11x + 30$. 5

20. Expand $(3x + 3y)^2$ and find its value when $x = 1$ and $y = 1$. 5

OR

Factorise $x^2 + 10x + 21$. 5

END OF QUESTION PAPER — Total: 40 Marks

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